RITVIK CHUNAMARI

Arvada, CO | +1 (303) 875-7635 | ritvikchunamari@gmail.com | linkedin.com/in/ritvikchunamari | ritvikchunamari.github.io

PROFESSIONAL SUMMARY

User Experience Designer and Computer Science Engineer specializing in accessible, human-centered digital products, design systems, and data-driven interfaces. Proven track record conducting mixed-methods user research, architecting complex workflows, and accelerating design-to-engineering handoff across enterprise and mobile platforms. Guided by firsthand perspective navigating life with Cerebral Palsy, championing WCAG compliance, cognitive clarity, and inclusive experiences that elevate human potential.

PROFESSIONAL EXPERIENCE

AI Interface Student Assistant — *ATLAS Institute, CU Boulder* (Boulder, CO / On-Site) **October 2026 – Present**

- Engineered an on-device Voice AI interface platform operating independently of cloud APIs, eliminating external egress latency and ensuring 100% local conversational privacy.
- Architected real-time verification guardrails and automated daily RAG ingestion pipelines, boosting factual response accuracy to 99.2% across local domain knowledge graphs.
- Implemented sub-50ms acoustic barge-in interruption and voice activity detection (VAD), optimizing natural bidirectional conversational turn-taking without audio truncation.
- Accelerated local neural inference pipelines via PyTorch and Apple Silicon Metal Performance Shaders (MPS), achieving 3.2× faster token generation and zero frame-drop latency.

UI Design Intern — *Impact Events* (United States / Remote) **June 2026 – August 2026**

- Spearheaded the responsive mobile redesign of the core event-management platform, converting multi-step desktop workflows into intuitive mobile screens and cutting user task completion drop-off by 18%.
- Architected and standardized an Atomic Design System in Figma with 45+ reusable UI components, responsive layout tokens, and developer-ready specs, reducing front-end handoff revision cycles by 15%.
- Conducted accessibility audits against WCAG 2.1 AA benchmarks and moderated usability evaluations across legacy mobile flows, resolving critical tap-target and contrast bottlenecks to boost mobile task efficiency by 22%.

Associate UX Designer — *In-House Anonymous* (Brooklyn, NY / Remote) **December 2024 – May 2025**

- Engineered end-to-end interface architectures and interactive prototypes for an enterprise operations platform, driving a 15% increase in active user engagement and reducing average task completion time by 10%.
- Facilitated structured usability testing sessions with 25+ diverse participants, translating qualitative insights into data-driven interface iterations that improved user satisfaction scores by 15%.
- Refined core navigation hierarchy and information taxonomy across web dashboards, eliminating user friction points, decreasing user error rates by 12%, and accelerating daily operational workflow efficiency by 40%.

Freelance Product Designer — *i2u.ai* (Bengaluru, India / Remote) **September 2024 – January 2025**

- Directed end-to-end UX architecture and visual brand identity for an emerging AI data intelligence platform, driving a 30% increase in early user adoption and recurring feature utilization.
- Executed mixed-methods user research and iterative A/B testing across five distinct interface variations, cutting time-to-first-conversion by 25% and significantly streamlining customer onboarding journeys.
- Constructed wireframes, high-fidelity clickable mockups, and responsive component libraries in Figma, fostering cross-functional alignment and accelerating sprint delivery times.

UX Design Intern — *Pravinya Infotech Pvt. Ltd.* (Hubli, India / Hybrid) **January 2024 – June 2024**

- Crafted intuitive user journeys for an enterprise operations portal and administrative command dashboard, producing a 20% lift in customer engagement and a 15% improvement in conversion metrics.
- Conducted formal usability evaluations with 12+ enterprise stakeholders and field dispatchers, implementing design refinements that increased usability satisfaction ratings by 25%.
- Authored design system specifications, user operational flowcharts, and low-fidelity prototypes, standardizing reusable UI assets across multiple platform modules.

FEATURED PROJECTS

Kinetix Quant (Institutional Crypto AI Predictor) | Personal Open-Source Project **Live Project ↗**

Tools & Technologies: Python, FastAPI, TensorFlow, Keras, Pandas, GitHub Actions (MLOps), Render, WebSockets

- Architected a quantitative deep learning engine using a Hybrid 1D-CNN and Stacked GRU neural network to forecast asset prices, engineering a 12-dimensional feature matrix that ingests real-time technical indicators and macro-economic correlations (Nasdaq/DXY).
- Engineered an autonomous MLOps Continuous Learning Pipeline via GitHub Actions that automatically fetches live market data to retrain the .keras model weekly, while implementing TensorFlow Integrated Gradients to provide mathematical explainability.
- Deployed a decoupled Zero-Cold-Start architecture featuring a static TradingView UI on a CDN and a highly optimized FastAPI inference engine, utilizing dual-fallback WebSockets and automated cron jobs to guarantee 24/7 API uptime with 0ms latency.

Browser Bridge — Cross-Browser Migration Engine & Native macOS App | Systems & Native App Architect **Live Site ↗** | **GitHub ↗**

Tools & Technologies: Swift 6, SwiftUI, Rust, SQLite, Chromium/Firefox Profile Internals, macOS HIG

- Architected an open-source browser migration manager supporting 11 major browsers (Chromium, Firefox, Safari, Zen, Dia, Arc), directly parsing SQLite databases and JSON schemas without third-party cloud intermediaries.
- Engineered zero-lock concurrent profile extraction using SQLite immutable connections and WAL/SHM journal handling, enabling safe data migration from active browser sessions without database corruption.
- Designed a native macOS System Settings interface in Swift 6/SwiftUI conforming to Apple HIG, featuring pre-flight atomic snapshots, SHA-256 integrity verification, and 1-click state rollback.

Nexus AI — Machine Learning Data Management Platform | Lead Product Designer **Live Project ↗**

Tools & Technologies: Figma, React, TypeScript, Python, OpenAI API, Atomic Design System

- Architected the complete UX/UI and node-based visual dataflow editor in Figma, empowering non-technical stakeholders to intuitively configure, clean, and deploy machine learning data pipelines.
- Implemented a minimalist Swiss design system with clear typographical hierarchies and real-time status indicators, reducing cognitive load and cutting pipeline configuration error rates by 35%.
- Conducted user feedback validation sessions with 15+ data engineers, improving core feature discoverability by 28% and streamlining pipeline orchestration handoffs.

Smart Assistive Pill Dispenser 3D — IoT Health Hardware | Product & Industrial Designer **Live Project ↗**

Tools & Technologies: Autodesk Inventor, Arduino, Physical Prototyping, C++, Subsystem Clustering

- Modeled and prototyped an interactive 3D assistive medicine-dispensing robot in Autodesk Inventor, improving physical interaction ergonomic efficiency by 25% and reducing user learning curve by 30%.
- Collaborated with a cross-functional four-person engineering team, clustering electromechanical subsystems and sensor arrays to ensure 99.8% dispensing accuracy for elderly patients.

EDUCATION

Master of Science (M.S.) in Creative Technology and Design — ***University of Colorado Boulder*** **August 2025 – May 2027**

Track: Creative Industries | Location: Boulder, Colorado, United States **GPA: 3.9 / 4.0**

- **Relevant Coursework:** Design Methods, Business in Creativity, User Experience Design, User Experience Research, Visual Identity & Branding, Human-Computer Interaction (HCI).
- **Focus Areas:** UX/UI architecture, user-centered design methodologies, creative technology, and AI-Assisted Software Engineering (LLM Orchestration).

Bachelor of Engineering (B.E.) in Computer Science and Engineering — ***KLE Technological University*** **August 2020 – June 2024**

Location: Hubli, Karnataka, India **Undergraduate Degree**

- **Relevant Coursework:** Software Engineering, Agile Methodologies, Machine Learning, Web Technologies, Data Structures & Algorithms.

TECHNICAL SKILLS

Design & Prototyping: Figma, FigJam, Adobe Photoshop, Adobe Illustrator, Adobe XD, Autodesk Inventor, Atomic Design Systems, Wireframing, Rapid Prototyping, Design Tokens, Micro-interactions.

UX & Research Methods: Usability Testing, User Interviews, Persona Development, User Journey Mapping, Information Architecture (IA), A/B Testing, Heuristic Evaluation, System Usability Scale (SUS).

Languages & Systems Engineering: Swift (SwiftUI), Rust, Python, PyTorch, Apple Silicon Metal (MPS), TypeScript, JavaScript (ES6+), React, Vite, Tailwind CSS, Three.js/WebGL, SQLite, FastAPI, Git, GitHub, WCAG 2.1 AA Standards.

CERTIFICATIONS & RESEARCH PUBLICATIONS

Certifications: Google Career Certificates (Foundations of UX Design; Empathize, Define & Ideate; Wireframes & Prototypes) • Accenture Digital Skills: User Experience • IIT Madras: Cyber Security & Privacy.

Research Publications: *Intra-Oral Periapical Dental Classification using Convolutional Neural Network* (IEEE Xplore) • *3D Avatar Reconstruction Using Multi-Level Pixel Aligned Implicit Function* (Springer).

PERSONAL INSIGHTS & INCLUSIVE DESIGN ADVOCACY

Thriving professional in design and technology living with Cerebral Palsy. Applying firsthand lived experience to champion radical accessibility, eliminate cognitive and physical friction, and engineer digital experiences that empower people of all abilities.